Introduction to Semi-Supervised Learning with CNN

Azadeh Alavi

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What is Semi-Supervised Learning?

- A machine learning approach using both labeled (X_l, Y_l) and unlabeled (X_u) data.
- Reduces labeling cost while maintaining accuracy.
- Useful when collecting labeled data is expensive or time-consuming, such as medical images.
- Works well for image recognition, speech processing, and Natural Language Processing (NLP) applications.

Why Do We Need Semi-Supervised Learning?

- Labeling images manually takes time—human annotators must label thousands of examples.
- Large datasets exist without labels—semi-supervised learning helps utilize them.
- Improves performance over purely supervised learning when labeled data is scarce.

Self-Training Method

1. Train a CNN classifier using labeled data.
2. Use the trained model to predict labels for unlabeled data.
3. Select only high-confidence predictions (avoiding uncertain labels).
4. Add confident predictions to the labeled set.
5. Retrain the CNN iteratively to improve accuracy.

Step 1: Train CNN on Labeled Data

run the provided code and observe

- The steps and how to execute them
- Mark down the explanation
- Change the hyperparameter and observe the impact
- How can this be applied to your Assignment-2, task-1?

Final Comparison

Does Semi-Supervised Learning Work?

- Fully Supervised CNN Accuracy: 83
- Semi-Supervised CNN Accuracy: 88
- Improvement due to utilizing unlabeled data for training.
- A great approach when labeled data is scarce!

Summary and References

- Semi-supervised learning can improve accuracy by leveraging unlabeled data.
- Self-training with CNNs involves iterative refinement with pseudo-labels.
- Future applications: Object detection, text classification, medical imaging.

References

Yarowsky, D. (1995). "Unsupervised Word Sense Disambiguation."

Scikit-learn documentation: `https://scikit-learn.org/stable/modules/semi_supervised.html`